```python
import pandas as pd

dataset=pd.read_csv("Placement.csv")

dataset
```

```
     sl_no gender  ssc_p    ssc_b  hsc_p    hsc_b     hsc_s  degree_p
\
0        1      M  67.00   Others  91.00   Others  Commerce     58.00

1        2      M  79.33  Central  78.33   Others   Science     77.48

2        3      M  65.00  Central  68.00  Central      Arts     64.00

3        4      M  56.00  Central  52.00  Central   Science     52.00

4        5      M  85.80  Central  73.60  Central  Commerce     73.30

..     ...    ...    ...      ...    ...      ...       ...       ...

210    211      M  80.60   Others  82.00   Others  Commerce     77.60

211    212      M  58.00   Others  60.00   Others   Science     72.00

212    213      M  67.00   Others  67.00   Others  Commerce     73.00

213    214      F  74.00   Others  66.00   Others  Commerce     58.00

214    215      M  62.00  Central  58.00   Others   Science     53.00


      degree_t workex  etest_p specialisation  mba_p      status
salary
0     Sci&Tech     No     55.0         Mkt&HR  58.80      Placed
270000.0
1     Sci&Tech    Yes     86.5        Mkt&Fin  66.28      Placed
200000.0
2    Comm&Mgmt     No     75.0        Mkt&Fin  57.80      Placed
250000.0
3     Sci&Tech     No     66.0         Mkt&HR  59.43  Not Placed
NaN
4    Comm&Mgmt     No     96.8        Mkt&Fin  55.50      Placed
425000.0
..         ...    ...      ...            ...    ...         ...
...
210  Comm&Mgmt     No     91.0        Mkt&Fin  74.49      Placed
400000.0
211   Sci&Tech     No     74.0        Mkt&Fin  53.62      Placed
275000.0
212  Comm&Mgmt    Yes     59.0        Mkt&Fin  69.72      Placed
295000.0
```

```
213  Comm&Mgmt     No     70.0         Mkt&HR  60.23      Placed
204000.0
214  Comm&Mgmt     No     89.0         Mkt&HR  60.22  Not Placed
NaN

[215 rows x 15 columns]
```

```
dataset.info() #Nan / Null = Nothing and 215 records, 148  non-null =
148 values remaining are null
#int64 = Whole Number without decimal point, object = Categorical
(text), float64 = decimal point included
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 215 entries, 0 to 214
Data columns (total 15 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   sl_no           215 non-null    int64
 1   gender          215 non-null    object
 2   ssc_p           215 non-null    float64
 3   ssc_b           215 non-null    object
 4   hsc_p           215 non-null    float64
 5   hsc_b           215 non-null    object
 6   hsc_s           215 non-null    object
 7   degree_p        215 non-null    float64
 8   degree_t        215 non-null    object
 9   workex          215 non-null    object
 10  etest_p         215 non-null    float64
 11  specialisation  215 non-null    object
 12  mba_p           215 non-null    float64
 13  status          215 non-null    object
 14  salary          148 non-null    float64
dtypes: float64(6), int64(1), object(8)
memory usage: 25.3+ KB
```

```
#Quantitative = Numerical data and #Qualitative = Categorical data

#Quan=["ssc_p","hsc_p","etest_p","degree_p","mba_p","salary"]
#Qual=["gender","ssc_b","hsc_b","degree_t"]
```

```
dataset.columns
```

```
Index(['sl_no', 'gender', 'ssc_p', 'ssc_b', 'hsc_p', 'hsc_b', 'hsc_s',
       'degree_p', 'degree_t', 'workex', 'etest_p', 'specialisation',
'mba_p',
       'status', 'salary'],
      dtype='object')
```

```
dir(dataset)
```

```
['T',
 '_AXIS_LEN',
 '_AXIS_ORDERS',
 '_AXIS_TO_AXIS_NUMBER',
 '_HANDLED_TYPES',
 '__abs__',
 '__add__',
 '__and__',
 '__annotations__',
 '__array__',
 '__array_priority__',
 '__array_ufunc__',
 '__bool__',
 '__class__',
 '__contains__',
 '__copy__',
 '__dataframe__',
 '__dataframe_consortium_standard__',
 '__deepcopy__',
 '__delattr__',
 '__delitem__',
 '__dict__',
 '__dir__',
 '__divmod__',
 '__doc__',
 '__eq__',
 '__finalize__',
 '__floordiv__',
 '__format__',
 '__ge__',
 '__getattr__',
 '__getattribute__',
 '__getitem__',
 '__getstate__',
 '__gt__',
 '__hash__',
 '__iadd__',
 '__iand__',
 '__ifloordiv__',
 '__imod__',
 '__imul__',
 '__init__',
 '__init_subclass__',
 '__invert__',
 '__ior__',
 '__ipow__',
 '__isub__',
 '__iter__',
 '__itruediv__',
 '__ixor__',
```

```
 '__le__',
 '__len__',
 '__lt__',
 '__matmul__',
 '__mod__',
 '__module__',
 '__mul__',
 '__ne__',
 '__neg__',
 '__new__',
 '__nonzero__',
 '__or__',
 '__pandas_priority__',
 '__pos__',
 '__pow__',
 '__radd__',
 '__rand__',
 '__rdivmod__',
 '__reduce__',
 '__reduce_ex__',
 '__repr__',
 '__rfloordiv__',
 '__rmatmul__',
 '__rmod__',
 '__rmul__',
 '__ror__',
 '__round__',
 '__rpow__',
 '__rsub__',
 '__rtruediv__',
 '__rxor__',
 '__setattr__',
 '__setitem__',
 '__setstate__',
 '__sizeof__',
 '__str__',
 '__sub__',
 '__subclasshook__',
 '__truediv__',
 '__weakref__',
 '__xor__',
 '_accessors',
 '_accum_func',
 '_agg_examples_doc',
 '_agg_see_also_doc',
 '_align_for_op',
 '_align_frame',
 '_align_series',
 '_append',
 '_arith_method',
```

```
 '_arith_method_with_reindex',
 '_as_manager',
 '_attrs',
 '_box_col_values',
 '_can_fast_transpose',
 '_check_inplace_and_allows_duplicate_labels',
 '_check_is_chained_assignment_possible',
 '_check_label_or_level_ambiguity',
 '_check_setitem_copy',
 '_clear_item_cache',
 '_clip_with_one_bound',
 '_clip_with_scalar',
 '_cmp_method',
 '_combine_frame',
 '_consolidate',
 '_consolidate_inplace',
 '_construct_axes_dict',
 '_construct_result',
 '_constructor',
 '_constructor_from_mgr',
 '_constructor_sliced',
 '_constructor_sliced_from_mgr',
 '_create_data_for_split_and_tight_to_dict',
 '_data',
 '_deprecate_downcast',
 '_dir_additions',
 '_dir_deletions',
 '_dispatch_frame_op',
 '_drop_axis',
 '_drop_labels_or_levels',
 '_ensure_valid_index',
 '_find_valid_index',
 '_flags',
 '_flex_arith_method',
 '_flex_cmp_method',
 '_from_arrays',
 '_from_mgr',
 '_get_agg_axis',
 '_get_axis',
 '_get_axis_name',
 '_get_axis_number',
 '_get_axis_resolvers',
 '_get_block_manager_axis',
 '_get_bool_data',
 '_get_cleaned_column_resolvers',
 '_get_column_array',
 '_get_index_resolvers',
 '_get_item_cache',
 '_get_label_or_level_values',
 '_get_numeric_data',
```

```
 '_get_value',
 '_getitem_bool_array',
 '_getitem_multilevel',
 '_getitem_nocopy',
 '_getitem_slice',
 '_gotitem',
 '_hidden_attrs',
 '_indexed_same',
 '_info_axis',
 '_info_axis_name',
 '_info_axis_number',
 '_info_repr',
 '_init_mgr',
 '_inplace_method',
 '_internal_names',
 '_internal_names_set',
 '_is_copy',
 '_is_homogeneous_type',
 '_is_label_or_level_reference',
 '_is_label_reference',
 '_is_level_reference',
 '_is_mixed_type',
 '_is_view',
 '_iset_item',
 '_iset_item_mgr',
 '_iset_not_inplace',
 '_item_cache',
 '_iter_column_arrays',
 '_ixs',
 '_logical_func',
 '_logical_method',
 '_maybe_align_series_as_frame',
 '_maybe_cache_changed',
 '_maybe_update_cacher',
 '_metadata',
 '_mgr',
 '_min_count_stat_function',
 '_needs_reindex_multi',
 '_pad_or_backfill',
 '_protect_consolidate',
 '_reduce',
 '_reduce_axis1',
 '_reindex_axes',
 '_reindex_multi',
 '_reindex_with_indexers',
 '_rename',
 '_replace_columnwise',
 '_repr_data_resource_',
 '_repr_fits_horizontal_',
 '_repr_fits_vertical_',
```

```
 '_repr_html_',
 '_repr_latex_',
 '_reset_cache',
 '_reset_cacher',
 '_sanitize_column',
 '_series',
 '_set_axis',
 '_set_axis_name',
 '_set_axis_nocheck',
 '_set_is_copy',
 '_set_item',
 '_set_item_frame_value',
 '_set_item_mgr',
 '_set_value',
 '_setitem_array',
 '_setitem_frame',
 '_setitem_slice',
 '_shift_with_freq',
 '_should_reindex_frame_op',
 '_slice',
 '_sliced_from_mgr',
 '_stat_function',
 '_stat_function_ddof',
 '_take_with_is_copy',
 '_to_dict_of_blocks',
 '_to_latex_via_styler',
 '_typ',
 '_update_inplace',
 '_validate_dtype',
 '_values',
 '_where',
 'abs',
 'add',
 'add_prefix',
 'add_suffix',
 'agg',
 'aggregate',
 'align',
 'all',
 'any',
 'apply',
 'applymap',
 'asfreq',
 'asof',
 'assign',
 'astype',
 'at',
 'at_time',
 'attrs',
 'axes',
```

```
 'backfill',
 'between_time',
 'bfill',
 'bool',
 'boxplot',
 'clip',
 'columns',
 'combine',
 'combine_first',
 'compare',
 'convert_dtypes',
 'copy',
 'corr',
 'corrwith',
 'count',
 'cov',
 'cummax',
 'cummin',
 'cumprod',
 'cumsum',
 'degree_p',
 'degree_t',
 'describe',
 'diff',
 'div',
 'divide',
 'dot',
 'drop',
 'drop_duplicates',
 'droplevel',
 'dropna',
 'dtypes',
 'duplicated',
 'empty',
 'eq',
 'equals',
 'etest_p',
 'eval',
 'ewm',
 'expanding',
 'explode',
 'ffill',
 'fillna',
 'filter',
 'first',
 'first_valid_index',
 'flags',
 'floordiv',
 'from_dict',
 'from_records',
```

```
 'ge',
 'gender',
 'get',
 'groupby',
 'gt',
 'head',
 'hist',
 'hsc_b',
 'hsc_p',
 'hsc_s',
 'iat',
 'idxmax',
 'idxmin',
 'iloc',
 'index',
 'infer_objects',
 'info',
 'insert',
 'interpolate',
 'isetitem',
 'isin',
 'isna',
 'isnull',
 'items',
 'iterrows',
 'itertuples',
 'join',
 'keys',
 'kurt',
 'kurtosis',
 'last',
 'last_valid_index',
 'le',
 'loc',
 'lt',
 'map',
 'mask',
 'max',
 'mba_p',
 'mean',
 'median',
 'melt',
 'memory_usage',
 'merge',
 'min',
 'mod',
 'mode',
 'mul',
 'multiply',
 'ndim',
```

```
 'ne',
 'nlargest',
 'notna',
 'notnull',
 'nsmallest',
 'nunique',
 'pad',
 'pct_change',
 'pipe',
 'pivot',
 'pivot_table',
 'plot',
 'pop',
 'pow',
 'prod',
 'product',
 'quantile',
 'query',
 'radd',
 'rank',
 'rdiv',
 'reindex',
 'reindex_like',
 'rename',
 'rename_axis',
 'reorder_levels',
 'replace',
 'resample',
 'reset_index',
 'rfloordiv',
 'rmod',
 'rmul',
 'rolling',
 'round',
 'rpow',
 'rsub',
 'rtruediv',
 'salary',
 'sample',
 'select_dtypes',
 'sem',
 'set_axis',
 'set_flags',
 'set_index',
 'shape',
 'shift',
 'size',
 'skew',
 'sl_no',
 'sort_index',
```

```
 'sort_values',
 'specialisation',
 'squeeze',
 'ssc_b',
 'ssc_p',
 'stack',
 'status',
 'std',
 'style',
 'sub',
 'subtract',
 'sum',
 'swapaxes',
 'swaplevel',
 'tail',
 'take',
 'to_clipboard',
 'to_csv',
 'to_dict',
 'to_excel',
 'to_feather',
 'to_gbq',
 'to_hdf',
 'to_html',
 'to_json',
 'to_latex',
 'to_markdown',
 'to_numpy',
 'to_orc',
 'to_parquet',
 'to_period',
 'to_pickle',
 'to_records',
 'to_sql',
 'to_stata',
 'to_string',
 'to_timestamp',
 'to_xarray',
 'to_xml',
 'transform',
 'transpose',
 'truediv',
 'truncate',
 'tz_convert',
 'tz_localize',
 'unstack',
 'update',
 'value_counts',
 'values',
 'var',
```

```
 'where',
 'workex',
 'xs']
```

```
dataset["ssc_p"].dtypes
```

```
dtype('float64')
```

```
dataset.dtypes
```

```
sl_no                int64
gender              object
ssc_p              float64
ssc_b               object
hsc_p              float64
hsc_b               object
hsc_s               object
degree_p           float64
degree_t            object
workex              object
etest_p            float64
specialisation      object
mba_p              float64
status              object
salary             float64
dtype: object
```

```
for columnName in dataset.columns:
    print(columnName)
```

```
sl_no
gender
ssc_p
ssc_b
hsc_p
hsc_b
hsc_s
degree_p
degree_t
workex
etest_p
specialisation
mba_p
status
salary
```

```
if(dataset["ssc_p"].dtype=='O'):
    print("qual")
else:
    print("quan")
```

```
quan
```

```
quan=[]
qual=[]
for columnName in dataset.columns:
    #print(columnName)
    if(dataset[columnName].dtype=='O'):
        #print("qual")
        qual.append(columnName)
    else:
        #print("quan")
        quan.append(columnName)
```

```
quan
```

```
['sl_no', 'ssc_p', 'hsc_p', 'degree_p', 'etest_p', 'mba_p', 'salary']
```

```
qual
```

```
['gender',
 'ssc_b',
 'hsc_b',
 'hsc_s',
 'degree_t',
 'workex',
 'specialisation',
 'status']
```

```
def quanQual(dataset):
    quan=[]
    qual=[]
    for columnName in dataset.columns:
        #print(columnName)
        if(dataset[columnName].dtype=='O'):
            #print("qual")
            qual.append(columnName)
        else:
            #print("quan")
            quan.append(columnName)
    return quan,qual
```

```
quanQual(dataset)
```

```
(['sl_no', 'ssc_p', 'hsc_p', 'degree_p', 'etest_p', 'mba_p',
'salary'],
 ['gender',
  'ssc_b',
```

```
  'hsc_b',
  'hsc_s',
  'degree_t',
  'workex',
  'specialisation',
  'status'])
```

```
quan,qual=quanQual(dataset)
```

```
quan
```

```
['sl_no', 'ssc_p', 'hsc_p', 'degree_p', 'etest_p', 'mba_p', 'salary']
```

```
qual
```

```
['gender',
 'ssc_b',
 'hsc_b',
 'hsc_s',
 'degree_t',
 'workex',
 'specialisation',
 'status']
```

```
dataset
```

```
     sl_no gender  ssc_p    ssc_b  hsc_p    hsc_b     hsc_s  degree_p
\
0        1      M  67.00   Others  91.00   Others  Commerce     58.00

1        2      M  79.33  Central  78.33   Others   Science     77.48

2        3      M  65.00  Central  68.00  Central      Arts     64.00

3        4      M  56.00  Central  52.00  Central   Science     52.00

4        5      M  85.80  Central  73.60  Central  Commerce     73.30

..     ...    ...    ...      ...    ...      ...       ...       ...

210    211      M  80.60   Others  82.00   Others  Commerce     77.60

211    212      M  58.00   Others  60.00   Others   Science     72.00

212    213      M  67.00   Others  67.00   Others  Commerce     73.00

213    214      F  74.00   Others  66.00   Others  Commerce     58.00

214    215      M  62.00  Central  58.00   Others   Science     53.00


      degree_t workex  etest_p specialisation  mba_p      status
```

```
salary
0     Sci&Tech     No     55.0         Mkt&HR  58.80      Placed
270000.0
1     Sci&Tech    Yes     86.5        Mkt&Fin  66.28      Placed
200000.0
2    Comm&Mgmt     No     75.0        Mkt&Fin  57.80      Placed
250000.0
3     Sci&Tech     No     66.0         Mkt&HR  59.43  Not Placed
NaN
4    Comm&Mgmt     No     96.8        Mkt&Fin  55.50      Placed
425000.0
..          ...    ...      ...            ...    ...         ...
...
210  Comm&Mgmt     No     91.0        Mkt&Fin  74.49      Placed
400000.0
211   Sci&Tech     No     74.0        Mkt&Fin  53.62      Placed
275000.0
212  Comm&Mgmt    Yes     59.0        Mkt&Fin  69.72      Placed
295000.0
213  Comm&Mgmt     No     70.0         Mkt&HR  60.23      Placed
204000.0
214  Comm&Mgmt     No     89.0         Mkt&HR  60.22  Not Placed
NaN

[215 rows x 15 columns]
```

Central Tendency

```
dataset["ssc_p"].mean()
```

```
67.30339534883721
```

```
dataset["ssc_p"].median()
```

```
67.0
```

```
dataset["ssc_p"].mode()
```

```
0    62.0
Name: ssc_p, dtype: float64
```

```
dataset["ssc_p"].mode(0)
```

```
0    62.0
Name: ssc_p, dtype: float64
```

```
dataset["ssc_p"].mode()[0]
```

```
62.0
```

```
dataset.describe()
```

```
            sl_no       ssc_p       hsc_p    degree_p     etest_p
mba_p  \
count  215.000000  215.000000  215.000000  215.000000  215.000000
215.000000
mean   108.000000   67.303395   66.333163   66.370186   72.100558
62.278186
std     62.209324   10.827205   10.897509    7.358743   13.275956
5.833385
min      1.000000   40.890000   37.000000   50.000000   50.000000
51.210000
25%     54.500000   60.600000   60.900000   61.000000   60.000000
57.945000
50%    108.000000   67.000000   65.000000   66.000000   71.000000
62.000000
75%    161.500000   75.700000   73.000000   72.000000   83.500000
66.255000
max    215.000000   89.400000   97.700000   91.000000   98.000000
77.890000
```

```
              salary
count     148.000000
mean   288655.405405
std     93457.452420
min    200000.000000
25%    240000.000000
50%    265000.000000
75%    300000.000000
max    940000.000000
```

```
descriptive=pd.DataFrame()
```

```
descriptive=pd.DataFrame(index=["Mean","Median","Mode"],columns=quan)
```

```
descriptive
```

```
       sl_no ssc_p hsc_p degree_p etest_p mba_p salary
Mean     NaN   NaN   NaN      NaN     NaN   NaN    NaN
Median   NaN   NaN   NaN      NaN     NaN   NaN    NaN
Mode     NaN   NaN   NaN      NaN     NaN   NaN    NaN
```

```
descriptive["hsc_p"]["Median"]
```

```
nan
```

```
descriptive["hsc_p"]["Median"]=0
```

```
descriptive
```

```
       sl_no ssc_p hsc_p degree_p etest_p mba_p salary
Mean     NaN   NaN   NaN      NaN     NaN   NaN    NaN
```

```
Median   NaN   NaN     0      NaN     NaN   NaN    NaN
Mode     NaN   NaN   NaN      NaN     NaN   NaN    NaN
```

```
descriptive=pd.DataFrame(index=["Mean","Median","Mode"],columns=quan)
for columnName in quan:
    descriptive[columnName]["Mean"]=dataset[columnName].mean()

descriptive
```

```
        sl_no      ssc_p      hsc_p   degree_p    etest_p
mba_p  \
Mean    108.0  67.303395  66.333163  66.370186  72.100558  62.278186

Median    NaN        NaN        NaN        NaN        NaN        NaN

Mode      NaN        NaN        NaN        NaN        NaN        NaN


                salary
Mean    288655.405405
Median            NaN
Mode              NaN
```

```
descriptive=pd.DataFrame(index=["Mean","Median","Mode"],columns=quan)
for columnName in quan:
    descriptive[columnName]["Mean"]=dataset[columnName].mean()
    descriptive[columnName]["Median"]=dataset[columnName].median()

descriptive
```

```
        sl_no      ssc_p      hsc_p   degree_p    etest_p
mba_p  \
Mean    108.0  67.303395  66.333163  66.370186  72.100558  62.278186

Median  108.0       67.0       65.0       66.0       71.0       62.0

Mode      NaN        NaN        NaN        NaN        NaN        NaN


                salary
Mean    288655.405405
Median       265000.0
Mode              NaN
```

```
descriptive=pd.DataFrame(index=["Mean","Median","Mode"],columns=quan)
for columnName in quan:
    descriptive[columnName]["Mean"]=dataset[columnName].mean()
    descriptive[columnName]["Median"]=dataset[columnName].median()
    descriptive[columnName]["Mode"]=dataset[columnName].mode()[0]

descriptive
```

```
        sl_no      ssc_p      hsc_p   degree_p    etest_p
mba_p  \
Mean    108.0  67.303395  66.333163  66.370186  72.100558  62.278186

Median  108.0       67.0       65.0       66.0       71.0       62.0

Mode        1       62.0       63.0       65.0       60.0       56.7


                salary
Mean    288655.405405
Median       265000.0
Mode         300000.0
```

Percentile

```
import numpy as np

np.percentile(dataset["ssc_p"],10)

52.0

np.percentile(dataset["ssc_p"],15)

55.0

np.percentile(dataset["ssc_p"],50)

67.0

dataset

     sl_no gender  ssc_p    ssc_b  hsc_p    hsc_b     hsc_s  degree_p
\
0        1      M  67.00   Others  91.00   Others  Commerce     58.00

1        2      M  79.33  Central  78.33   Others   Science     77.48

2        3      M  65.00  Central  68.00  Central      Arts     64.00

3        4      M  56.00  Central  52.00  Central   Science     52.00

4        5      M  85.80  Central  73.60  Central  Commerce     73.30

..     ...    ...    ...      ...    ...      ...       ...       ...

210    211      M  80.60   Others  82.00   Others  Commerce     77.60

211    212      M  58.00   Others  60.00   Others   Science     72.00
```

```
212    213      M  67.00   Others  67.00   Others  Commerce     73.00

213    214      F  74.00   Others  66.00   Others  Commerce     58.00

214    215      M  62.00  Central  58.00   Others   Science     53.00


      degree_t workex  etest_p specialisation  mba_p      status
salary
0     Sci&Tech     No     55.0         Mkt&HR  58.80      Placed
270000.0
1     Sci&Tech    Yes     86.5        Mkt&Fin  66.28      Placed
200000.0
2    Comm&Mgmt     No     75.0        Mkt&Fin  57.80      Placed
250000.0
3     Sci&Tech     No     66.0         Mkt&HR  59.43  Not Placed
NaN
4    Comm&Mgmt     No     96.8        Mkt&Fin  55.50      Placed
425000.0
..          ...    ...      ...            ...    ...         ...
...
210  Comm&Mgmt     No     91.0        Mkt&Fin  74.49      Placed
400000.0
211   Sci&Tech     No     74.0        Mkt&Fin  53.62      Placed
275000.0
212  Comm&Mgmt    Yes     59.0        Mkt&Fin  69.72      Placed
295000.0
213  Comm&Mgmt     No     70.0         Mkt&HR  60.23      Placed
204000.0
214  Comm&Mgmt     No     89.0         Mkt&HR  60.22  Not Placed
NaN

[215 rows x 15 columns]
```

```
np.percentile(dataset["ssc_p"],75)
```

```
75.7
```

```
np.percentile(dataset["ssc_p"],99)
```

```
87.0
```

```
np.percentile(dataset["ssc_p"],100)
```

```
89.4
```

```
descriptive=pd.DataFrame(index=["Mean","Median","Mode","Q1:25%","Q2:50
%","Q3:75%","Q4:100%"],columns=quan)
for columnName in quan:
    descriptive[columnName]["Mean"]=dataset[columnName].mean()
    descriptive[columnName]["Median"]=dataset[columnName].median()
```

```
    descriptive[columnName]["Mode"]=dataset[columnName].mode()[0]


descriptive
```

```
         sl_no      ssc_p      hsc_p   degree_p    etest_p      mba_p
\
Mean     108.0  67.303395  66.333163  66.370186  72.100558  62.278186

Median   108.0       67.0       65.0       66.0       71.0       62.0

Mode         1       62.0       63.0       65.0       60.0       56.7

Q1:25%     NaN        NaN        NaN        NaN        NaN        NaN

Q2:50%     NaN        NaN        NaN        NaN        NaN        NaN

Q3:75%     NaN        NaN        NaN        NaN        NaN        NaN

Q4:100%    NaN        NaN        NaN        NaN        NaN        NaN


                 salary
Mean     288655.405405
Median        265000.0
Mode          300000.0
Q1:25%             NaN
Q2:50%             NaN
Q3:75%             NaN
Q4:100%            NaN
```

```
descriptive=pd.DataFrame(index=["Mean","Median","Mode","Q1:25%","Q2:50
%","Q3:75%","Q4:100%"],columns=quan)
for columnName in quan:
    descriptive[columnName]["Mean"]=dataset[columnName].mean()
    descriptive[columnName]["Median"]=dataset[columnName].median()
    descriptive[columnName]["Mode"]=dataset[columnName].mode()[0]
    descriptive[columnName]
["Q1:25%"]=np.percentile(dataset[columnName],25)


descriptive
```

```
         sl_no      ssc_p      hsc_p   degree_p    etest_p      mba_p
\
Mean     108.0  67.303395  66.333163  66.370186  72.100558  62.278186

Median   108.0       67.0       65.0       66.0       71.0       62.0

Mode         1       62.0       63.0       65.0       60.0       56.7
```

```
Q1:25%    54.5       60.6       60.9       61.0       60.0     57.945

Q2:50%     NaN        NaN        NaN        NaN        NaN        NaN

Q3:75%     NaN        NaN        NaN        NaN        NaN        NaN

Q4:100%    NaN        NaN        NaN        NaN        NaN        NaN


                 salary
Mean     288655.405405
Median        265000.0
Mode          300000.0
Q1:25%             NaN
Q2:50%             NaN
Q3:75%             NaN
Q4:100%            NaN
```

```
descriptive=pd.DataFrame(index=["Mean","Median","Mode","Q1:25%","Q2:50
%","Q3:75%","Q4:100%"],columns=quan)
for columnName in quan:
    descriptive[columnName]["Mean"]=dataset[columnName].mean()
    descriptive[columnName]["Median"]=dataset[columnName].median()
    descriptive[columnName]["Mode"]=dataset[columnName].mode()[0]
    descriptive[columnName]["Q1:25%"]=dataset.describe()[columnName]
["25%"]
```

```
descriptive
```

```
         sl_no      ssc_p      hsc_p   degree_p    etest_p      mba_p
\
Mean     108.0  67.303395  66.333163  66.370186  72.100558  62.278186

Median   108.0       67.0       65.0       66.0       71.0       62.0

Mode         1       62.0       63.0       65.0       60.0       56.7

Q1:25%    54.5       60.6       60.9       61.0       60.0     57.945

Q2:50%     NaN        NaN        NaN        NaN        NaN        NaN

Q3:75%     NaN        NaN        NaN        NaN        NaN        NaN

Q4:100%    NaN        NaN        NaN        NaN        NaN        NaN


                 salary
Mean     288655.405405
Median        265000.0
Mode          300000.0
Q1:25%        240000.0
```

```
Q2:50%              NaN
Q3:75%              NaN
Q4:100%             NaN
```

```
descriptive=pd.DataFrame(index=["Mean","Median","Mode","Q1:25%","Q2:50
%","Q3:75%","Q4:100%"],columns=quan)
for columnName in quan:
    descriptive[columnName]["Mean"]=dataset[columnName].mean()
    descriptive[columnName]["Median"]=dataset[columnName].median()
    descriptive[columnName]["Mode"]=dataset[columnName].mode()[0]
    descriptive[columnName]["Q1:25%"]=dataset.describe()[columnName]
["25%"]
    descriptive[columnName]["Q2:50%"]=dataset.describe()[columnName]
["50%"]
    descriptive[columnName]["Q3:75%"]=dataset.describe()[columnName]
["75%"]
    descriptive[columnName]["Q4:100%"]=dataset.describe()[columnName]
["max"]
descriptive
```

```
         sl_no      ssc_p      hsc_p   degree_p    etest_p      mba_p
\
Mean     108.0  67.303395  66.333163  66.370186  72.100558  62.278186

Median   108.0       67.0       65.0       66.0       71.0       62.0

Mode         1       62.0       63.0       65.0       60.0       56.7

Q1:25%    54.5       60.6       60.9       61.0       60.0     57.945

Q2:50%   108.0       67.0       65.0       66.0       71.0       62.0

Q3:75%   161.5       75.7       73.0       72.0       83.5     66.255

Q4:100%  215.0       89.4       97.7       91.0       98.0      77.89


                 salary
Mean     288655.405405
Median        265000.0
Mode          300000.0
Q1:25%        240000.0
Q2:50%        265000.0
Q3:75%        300000.0
Q4:100%       940000.0
```

```
descriptive=pd.DataFrame(index=["Mean","Median","Mode","Q1:25%","Q2:50
%","Q3:75%","99%","Q4:100%"],columns=quan)
for columnName in quan:
    descriptive[columnName]["Mean"]=dataset[columnName].mean()
    descriptive[columnName]["Median"]=dataset[columnName].median()
```

```
    descriptive[columnName]["Mode"]=dataset[columnName].mode()[0]
    descriptive[columnName]["Q1:25%"]=dataset.describe()[columnName]
["25%"]
    descriptive[columnName]["Q2:50%"]=dataset.describe()[columnName]
["50%"]
    descriptive[columnName]["Q3:75%"]=dataset.describe()[columnName]
["75%"]
    descriptive[columnName]
["99%"]=np.percentile(dataset[columnName],99)
    descriptive[columnName]["Q4:100%"]=dataset.describe()[columnName]
["max"]
descriptive
```

	sl_no	ssc_p	hsc_p	degree_p	etest_p	mba_p
Mean	108.0	67.303395	66.333163	66.370186	72.100558	62.278186
Median	108.0	67.0	65.0	66.0	71.0	62.0
Mode	1	62.0	63.0	65.0	60.0	56.7
Q1:25%	54.5	60.6	60.9	61.0	60.0	57.945
Q2:50%	108.0	67.0	65.0	66.0	71.0	62.0
Q3:75%	161.5	75.7	73.0	72.0	83.5	66.255
99%	212.86	87.0	91.86	83.86	97.0	76.1142
Q4:100%	215.0	89.4	97.7	91.0	98.0	77.89

```
                salary
Mean     288655.405405
Median        265000.0
Mode          300000.0
Q1:25%        240000.0
Q2:50%        265000.0
Q3:75%        300000.0
99%                NaN
Q4:100%       940000.0
```